From Longitudinal Context to Safe Action: A Co-Versioned Governance Contract with Layer 1 Deterministic Safety Barrier for Primary Care AI

Nguyen Ngoc Thien, Trinh Minh Quang
Hai Ba Trung High School, Hue, Vietnam
Corresponding author: `kakangocthien109@gmail.com`

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Background. In primary care and family medicine, longitudinal electronic health records encompass multi-encounter clinical histories, chronic disease management trajectories, and sensitive personal health data. A persistent primary care AI system may read an authorized patient snapshot, but return a proposed clinical action or writeback only after the record, patient consent, or clinical policy has changed (time-of-check to time-of-use / TOCTOU drift). Furthermore, feeding complete unminimized health records to large language models (LLMs) risks severe Protected Health Information (PHI) over-disclosure and exposes clinicians and patients to LLM hallucinations during temporal arithmetic (such as preventive screening intervals and immunization due dates) and drug-drug interaction (DDI) evaluations.

Methods. We report the GLHS (Governed Longitudinal Health Snapshot) architecture implemented in CLARA-Care for family medicine and primary care workflows. GLHS introduces a **Dual-Layer State Barrier**: (1) a **Layer 1 Deterministic Reconciliation Kernel** (a trusted, non-LLM engine at the API gateway tier in Python/PostgreSQL) that deterministically evaluates bitemporal validity ($\tau_{\text{valid}} \cap \tau_{\text{known}}$), calculates protocol due-window arithmetic (preventive screening recall schedules, chronic care due dates, and immunization grace periods) with 100% precision and recall, checks severe drug-drug interactions, verifies dynamic patient consent and role permissions, and compiles a cryptographic Merkle digest (Π_{Merkle}) into a Task-Bounded Health State Snapshot (THSS) enforcing **Zero-PHI over-disclosure**; and (2) a **Layer 2 Epistemic Arbiter** (LLM) isolated from raw temporal math to focus strictly on qualitative synthesis. Model proposals retain the exact THSS binding to a Governed State Transition (GST) that re-verifies state and governance freshness before writeback. We evaluated GLHS using deterministic contract tests, PostgreSQL concurrency matrices ($N = 12$ schedules), a frozen 64-subject synthetic cohort across Claude Sonnet 4.6 and Gemini 3.6 Flash High (1,152 repeated model-condition evaluations), and a sealed 384-subject primary care cohort under Schuirmann's Two One-Sided Tests (TOST) biostatistical non-inferiority framework ($\alpha = 0.05$, prespecified clinical equivalence margin $\delta = \pm 0.020$).

Results. In the biostatistical TOST equivalence analysis ($N = 384$), Strict THSS demonstrated formal non-inferiority to full unminimized history in primary care clinical decision quality: lower TOST $t_1 = +5.3072$ ($p_1 = 9.44 \times 10^{-8}$), upper TOST $t_2 = -12.1114$ ($p_2 = 4.15 \times 10^{-29}$), overall $p_{\text{TOST}} = 9.44 \times 10^{-8} \ll 0.001$, with the 95% confidence interval $[-1.233\%, -0.330\%]$ fully bounded within $[-\delta, +\delta]$ (exact two-sided sign-test $p = 0.867$). Context minimization delivered an 87.4% prompt token reduction (mean 412 vs. 3,280 tokens), a 68.2% latency reduction, Zero-PHI over-disclosure (0.0% unintended sensitive entity exposure), and 100% elimination of read-to-write TOCTOU state corruptions ($p = 1.727 \times 10^{-77}$). The Layer 1 non-LLM safety kernel achieved 100% precision and recall on temporal due-window evaluations. In the 64-subject cohort, Strict THSS achieved all-axis exact match for 63/64 Claude subjects (98.44%) and 64/64 Gemini subjects (100%). In the 12-schedule PostgreSQL TOCTOU matrix, zero forbidden commits occurred across consent revocations, role transitions, policy updates, and concurrent state drifts. Bounded finite-state exploration verified 21,361 states and 90,432 transitions to depth 5 with zero invariant violations.

Conclusion. GLHS demonstrates that a Layer 1 non-LLM safety barrier and co-versioned disclosure-to-commit contract eliminate read-to-write state inconsistencies and guarantee Zero-PHI over-disclosure in primary care settings without compromising clinical decision accuracy, as proven by formal TOST biostatistical non-inferiority. Current evidence is derived from synthetic cohorts and software benchmarks; clinical deployment requires prospective real-world clinical validation.

Keywords: primary care informatics; family medicine AI; data minimization; Zero-PHI; biostatistical equivalence; TOST; non-LLM safety barrier; state consistency; clinical governance.